

Soybean Yield and Biomass Responses to Increasing Plant Population Among Diverse Maturity Groups: I. Agronomic Characteristics

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ABSTRACT

Soybean [*Glycine max* (L.) Merr.] production systems that require less than 100 d from sowing to maturity are useful for avoiding drought and decreasing irrigation requirement in areas with relatively long growing seasons (>150 d) and seasonal moisture limitations. Previous experiments have evaluated the response of either maturity group (MG) IV and earlier or MG V and later soybean to increased plant population, but there is a paucity of data quantifying population density responses across the entire range of MGs grown in these areas. Therefore, we evaluated the responses of MG 00, 0, I, II, III, IV, V, and VI soybean to populations of 10, 20, 40, 60, and 100 seeds m⁻² sown in 19-cm rows and irrigated as needed at Fayetteville, AR, in 2001, 2002, and 2003. The response of soybean yield to increased plant populations was described well by an exponential model that predicted an asymptotic yield plateau at high plant populations. Asymptotic yield was similar for MG I through VI cultivars, but plant population required to reach the asymptote generally decreased as soybean maturity lengthened. Harvest index (HI) values generally increased slightly with increased plant population in MG 00 and 0 soybean, decreased slightly with increasing plant populations in MG V and VI soybean, and had no response to increasing plant populations in MG I through IV soybean. Height of first fertile node increased as plant population increased and as soybean maturity lengthened. This research demonstrates that a broad range of soybean MGs can produce similar yield in the Midsouth, but optimal seeding densities and irrigation requirements vary by maturity. Further, this research demonstrates some of the difficulties that can be encountered when expressing soybean yield as an empirical function of soybean population density.

EARLY-PLANTED, short-season crops are useful for reducing irrigation requirement or for increasing rain-fed yield in the presence of late-season drought (Edwards et al., 2003; Oad and Azim, 2002; Purcell et al., 2003), and for these reasons they have been widely adopted in the midsouthern USA (Heatherly, 1999). Success of short-season soybean (MG ≤ IV) production in the Midsouth is contingent on higher population and more narrow rows than those for full-season crops whether sown in early spring to avoid late-summer drought (Heatherly, 1999) or when sown later in a double-cropping system of soybean following winter wheat (*Triticum aestivum* L.) (Ball et al., 2000). Therefore, plant population response data help producers make better-

informed decisions concerning management of full-season and short-season crops.

The majority of soybean plant population work to date has reflected the cultural practices and soybean maturities traditionally grown in the area where the research was conducted. As a result, previous experiments have included a broad range of row spacing and population densities, but few experiments have evaluated more than two MGs within an experiment (Boquet, 1990; Elmore, 1998; Ethredge et al., 1989; Holshouser and Jones, 2003; Parvez et al., 1989; Weber et al., 1966). This makes comparisons of optimal population density among different MGs difficult, as soybean responses to population density vary by environment. Furthermore, soybean production technology and trends have changed since many of these experiments were conducted. In the lower Mississippi Delta, for example, MG III and IV soybean cultivars are now planted on large portion of the hectareage formerly devoted to MG V and VI soybean (Heatherly, 1999). However, there is little information published on the response of short-season soybean to increased plant population in a southern environment, and current seeding density recommendations remain at approximately 25 seeds m⁻² (Ashlock et al., 2000).

The effect of row spacing on the amount of light intercepted by the soybean canopy is an integral part of the dynamic relationship between soybean yield and seeded population (Weber et al., 1966). While there have been several experiments evaluating optimal soybean seeded population in a narrow-row production system, the row spacing defined by the term "narrow row" has changed over time and varies greatly by geographic region. In the southern USA, for example, the term narrow-row spacing has been used to describe almost any row spacing less than 1 m. Planting equipment and row spacing, however, have changed considerably over the past 15 yr, with drilled-soybean (19 cm or less) becoming more popular in the southern USA, due to increased yield of short-season soybean associated with narrower row spacing (Ball et al., 2000) as well as increased competitiveness of soybean with weeds (Norsworthy and Oliver, 2002).

Changes in management practices and the blurring of traditional production areas for various MG soybean creates a need for more information on the effect of increased plant populations in a more southern environment. Our objectives in this research were to (i) determine the yield-optimizing plant population of MG 00 through VI soybean when seeded in a well-irrigated, narrow-row production system; (ii) evaluate the irrigation requirement for different MGs; and (iii) compare

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Abbreviations: DOY, day of year; HI, harvest index; MG, maturity group.

agronomic characteristics among MG 00 through VI soybean cultivars when sown in the Midsouth.

MATERIALS AND METHODS

Field studies were conducted in 2001, 2002, and 2003 at Fayetteville, AR (36°05' N, 94°10' W) on a Captina silt loam (fine-silty, siliceous, active, mesic Typic Fragiudult). Experimental design was a split-plot arrangement of treatments in a randomized complete block with four replications (blocks). In 2001 and 2003 main plots were soybean MGs (MG 00, Trail; MG 0, Lambert; MG I, IA 1006; MG II, IA 2008; MG III, Macon; MG IV, Pioneer 94B01; MG V, Hutcheson; and MG VI, NK 6262) and subplots were seeded population (10, 20, 40, 60, and 100 seeds m⁻²) randomized within MG. In 2002, there were two cultivars for each MG and main plots were soybean MG (00, 0, I, II, III, IV, V, and VI) and subplots were the interaction of seeded population (10, 20, 40, 60, and 100 seeds m⁻²) and soybean cultivar (MG 00, Jim and Trail; MG 0, AC Comoran and Lambert; MG I, IA 1006 and MN 1801; MG II, Dwight and IA 2008; MG III, Macon and Pana; MG IV, Pioneer 94B01 and MPV 437; MG V, Caviness and Hutcheson; and MG VI, Desha and NK 6262) randomized within MG. These cultivars were selected as well-adapted cultivars for the midsouthern production area based on previous research (Edwards et al., 2003; Ishibashi et al., 2003) and Arkansas Soybean Variety Performance Tests (Dombek et al., 2001).

Plots were prepared for seeding by applying 60 kg ha⁻¹ P₂O₅ and 80 kg ha⁻¹ K₂O and tilling to a 3-cm depth. Weeds were controlled by pre-emergence herbicide application and hand weeding. Insecticide applications were made as needed. Plots were 6 m long and seeding was performed using a cone-type conventional drill seeder with seven 19-cm rows. Plant population for each plot was determined by five separate counts of the number of emerged plants in 1 m of row, approximately 1 week after emergence. No reduction in plant population density was observed after the initial stand count, but some self-thinning may have decreased the surviving population at maturity (Ethredge et al., 1989). Actual plant populations after emergence deviated slightly from seeded population; therefore, plant population, rather than seeded population, was used for all analyses. Soybean developmental stages (Fehr and Caviness, 1977) were determined each year by evaluating soybean from each MG on an approximate 3-d interval following soybean emergence.

Our target sowing date was mid-May, which is a typical sowing date for the northern portion of the Midsouth but is later than sowing dates used in the early production system of the lower Midsouth (early to mid-April, Heatherly, 1999). In 2001 plots were initially seeded May 17, but a 4-cm rainfall immediately after sowing reduced emergence, so the entire plot area was sprayed with 1 kg a.i. ha⁻¹ glyphosate [N-(phosphonomethyl)glycine], tilled, and resown on June 10. Emergence was June 16. In 2002 plots were drill-seeded on May 7, and emergence was May 20. In 2003 plots were initially seeded on May 12, but cool, wet conditions and 7 cm of rainfall in the week following sowing reduced emergence (<70%), and the plot area was sprayed with 1 kg ha⁻¹ glyphosate, tilled, and resown on May 27. Emergence was 3 June 2003.

Irrigation was applied by overhead sprinklers when the estimated soil-water deficit reached 30 mm. The amount of irrigation varied from 13 to 30 mm per application depending on wind conditions. Soil-water deficit was estimated using the University of Arkansas' Irrigation Scheduling program, which is available for download (www.aragriculture.org/computer/schedule/default.asp; verified 14 Apr. 2005). This program subtracts daily estimates of crop evapotranspiration from daily inputs of either irrigation or rainfall (Cahoon et al., 1990).

Irrigation is recommended once the cumulative soil-water deficit reaches a critical value that is determined by soil characteristics and rooting depth. Although plant population may have affected evapotranspiration and soil-water deficit very early in the season (Reicosky et al., 1985), irrigation decisions were made based on estimated water deficit for the experiment as a whole. This may have overestimated the amount of irrigation required for lower plant populations, especially in MG II and earlier soybean. The inability to irrigate on a plot-by-plot basis, however, necessitated estimation of irrigation requirement based on the experiment as a whole.

To remove any edge effects, the two outside rows and 0.6 m from the plot ends were removed immediately before harvest. At this time, the distance from the soil surface to first fertile node was recorded. Harvest index samples were collected immediately before harvest by hand harvesting the above-ground portion of soybean from 1 m² of plot area. Samples were dried at 50°C for a period of 3 to 5 d, weighed, and threshed. Harvest index was calculated as the ratio of seed mass to total aboveground plant mass. Soybean was harvested using a small-plot combine and grain weights were corrected to 130 g kg⁻¹ moisture content.

The responses (Y) of yield (g m⁻²) and average height (cm) of first fertile node were modeled as an exponential function of plant population (x , plants m⁻²) using a nonlinear regression model where:

$$Y = \alpha(1 - e^{-\beta x}) \quad [1]$$

In Eq. [1], the coefficient α is the predicted, asymptotic maximum, and β represents the responsiveness of Y as plant population increased. In the context of this experiment, a smaller β indicates that a greater plant population was required to obtain maximum Y . Since Eq. [1] constrains the regressions to an intercept of 0, an R^2 value was calculated as (Ryan, 1997):

$$R^2 = 1 - (\text{Residual SS/Uncorrected total SS}) \quad [2]$$

Homogeneity of regression coefficients (C_1 and C_2) was determined using a Z test, with the null hypothesis being $C_1 - C_2 = 0$ (Kanji, 1993). We calculated Z values as:

$$Z = (C_1 - C_2)/(\text{SE}_1^2 + \text{SE}_2^2)^{0.5} \quad [3]$$

In Eq. [3], SE_1 and SE_2 refer to the standard errors associated with regression coefficients C_1 and C_2 , respectively. From this, the probability of the regression coefficients differing was determined as two times the Z value using a normal probability distribution.

Equation [1] has been successfully used in similar experiments evaluating the response of crop parameters to inputs that exhibit decreasing marginal return to increased input level, such as N fertilizer (Cerrato and Blackmer, 1990), the response of light interception to soybean to plant population (Purcell et al., 2002), and the yield response of maize (*Zea mays* L.) to population density (Edwards et al., 2005b; Ware et al., 1982). Furthermore, Eq. [1] (exponential rise to an asymptotic maximum) fit the overall trend of our yield and pod-height data well. First order regression provided no better fit (data not shown) for the majority of the yield and pod-height data and did not accurately represent the asymptotic nature of our data. Higher order terms were generally nonsignificant (data not shown) and did not improve the fitted model. Finally, fitting a second-order polynomial would indicate that yield declined at higher plant population densities, but this phenomenon was not observed in our experiment.

Harvest index response to plant population was analyzed by considering MG as a covariate. Experimental replication (block) and all of its interactions were treated as random effects. Maturity group was treated as a nominal variable and plant population was treated as a continuous variable in the same model. This analysis generated an intercept and slope

describing the relationship of the dependent and independent variables and is generally preferred to multiple comparison procedures or means separation when a stepwise series of treatments is used (Chew, 1976; Carmer and Walker, 1982; Peterson, 1977).

RESULTS AND DISCUSSION

Phenology and Irrigation

Plant population did not affect soybean maturity in any year of the study, and differences in phenology between cultivars within a MG were minor (<1 d); therefore, phenology data were averaged over soybean seeded population and cultivar (Table 1). There were 54, 64, and 59 d difference between the earliest and latest MGs in time from emergence to R7 in 2001, 2002, and 2003, respectively. In 2001 most of the difference in overall maturity was associated with a 51-d difference among MGs in duration of vegetative growth (Emergence to R1) as compared with a 16-d difference in duration of reproductive growth (R1–R7). Differences among MGs in overall maturity were fairly evenly split between vegetative and reproductive growth (≈ 30 d each) in 2002 and 2003. The longer duration of vegetative growth in 2002 and 2003 as compared to that of 2001, was probably the result of earlier sowing and cooler temperatures, as it has been well documented (Sinclair et al., 1991) that phenology is dependent on both temperature and photoperiod (Major et al., 1975).

Irrigation requirements generally increased as soybean maturity increased, but differences in irrigation requirement among MGs varied by year (Table 1). For example, in 2001, MG III and later soybean required 50% more irrigation than MG 00 soybean and 25% more irrigation than MG I and II soybean. The Early Soybean Production System in the Midsouth (Heath-

erly, 1999) utilizes early sowing dates and MG III and IV cultivars to avoid significant drought stress and to decrease irrigation requirements by maturing before the risk of drought is greatest (Purcell et al., 2003). Earlier planting in our experiment did not, however, always result in reduced irrigation requirement (Table 1). For example MG III and IV soybean seeded on day of year (DOY) 167 in 2001 required less irrigation than the same MG soybean sown on DOY 140 in 2002 and DOY 154 in 2003. This was due to differences in rainfall pattern among years of the experiment, as there was much greater late-season rainfall in 2001 than in subsequent years (Fig. 1A, 1B, 1C). Nevertheless, while early sowing did not guarantee reduced irrigation requirement, historical data indicate that risk of drought stress is greatly reduced by early sowing (Purcell et al., 2003), and our data

Table 1. Dates of emergence and phenological events and the amount of irrigation applied from emergence to R6 for MG 00 through VI soybean at Fayetteville, AR, in 2001, 2002, and 2003.

Year	Maturity group	Emergence	R1	R5	R6	R7	Irrigation
		Day of year	– Days after emergence –				cm
2001	00	167	15	41	55	68	10
	0	167	20	51	59	73	11
	I	167	24	55	66	84	12
	II	167	27	60	72	94	12
	III	167	31	66	81	96	15
	IV	167	34	66	85	101	15
2002	V	167	61	86	113	122	15
	VI	167	66	91	113	122	15
	00	140	27	47	58	65	9
	0	140	30	52	65	70	11
	I	140	30	56	73	78	14
	II	140	31	60	77	90	14
2003	III	140	35	62	84	99	17
	IV	140	40	71	91	105	18
	V	140	58	91	118	124	27
	VI	140	60	93	121	129	27
	00	154	27	45	62	74	14
	0	154	28	47	73	77	16
	I	154	31	55	78	86	21
	II	154	33	62	84	92	23
	III	154	34	65	92	102	23
	IV	154	42	69	95	109	23
	V	154	53	81	125	133	27
	VI	154	55	84	125	133	27

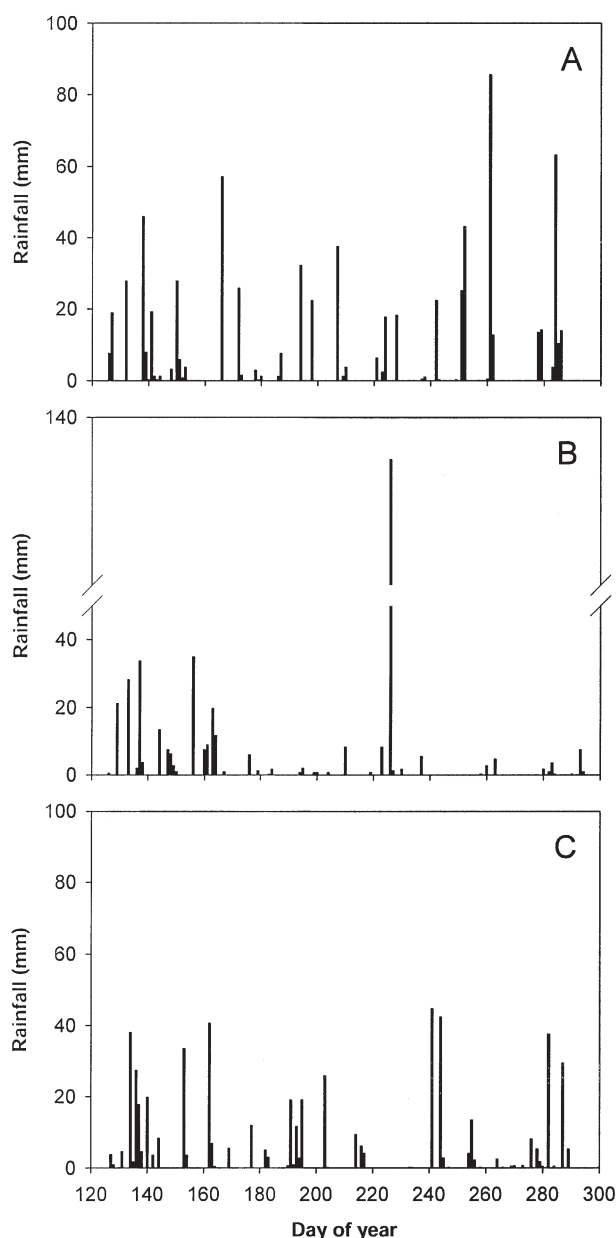


Fig. 1. Daily precipitation for day of year 125 through 300 in (A) 2001, (B) 2002, and (C) 2003 at Fayetteville, AR.

clearly indicate that shorter-season cultivars reduce irrigation requirement due to a shorter duration of growth, as compared to that to longer-season cultivars.

Yield as a Function of Plant Population

With few exceptions, Eq. [1] described the relationship between yield and plant population well for all MGs with R^2 values ranging from 0.81 to 0.99. Soybean yield increased more gradually in response to increased plant population for early-maturing cultivars than for later-maturing cultivars (Fig. 2), which is indicated by a smaller value of the exponential coefficient (β). For example, at high values of plant population, asymptotic yield of 395 g m^{-2} (MG 0) and 488 g m^{-2} were predicted for MG 0 and IV soybean, respectively. Data from all years and MGs followed the general response illustrated in Fig. 2; therefore, only coefficients, standard errors, and significance level for each regression are shown (Table 2).

For MG VI in 2001, MG III through VI in 2002, and MG V and VI in 2003, the nonlinear regression for yield failed to converge and predict β coefficient. In these cases, the asymptote was reached at populations $< 10 \text{ plants m}^{-2}$, and data points, regardless of plant population, fell close to the asymptote (α coefficient). The asymptotic relationship observed between soybean yield and plant population was likely the result of increased interplant competition which led to decreasing marginal increases in yield for added plant population (Egli, 1988; Duncan, 1986).

Pairwise statistical comparison of regression coefficients (data not shown) revealed that β coefficients within the same MG did not differ among years of the experiment ($P \leq 0.05$). However, α coefficients were statistically different in 28 of the 48 possible pairwise comparisons (six for each MG). This indicates that the relative response of yield to population density within a MG was similar among years but that the asymptotes (or yield potential) differed.

The lower asymptotic yield of MG 00 and 0 soybean

when sown before June 1 in the Midsouth was most evident in the 2002 growing season. The daily maximum temperature following soybean emergence in 2002 was often below 25°C . Cool, overcast growing conditions would have the greatest impact on the shortest season cultivars as soybean leaf expansion rate is temperature dependent (Sinclair, 1984). The reduced leaf expansion rate would limit the cumulative amount of intercepted photosynthetically active radiation and thereby reduce total biomass production and grain yield (Edwards et al., 2005a; Purcell et al., 2002).

It is notable that in 2003, yields for all MGs were greater than in 2001 and 2002 (Table 2). Rainfall distribution was generally better for crop production in 2003 than in 2002 but was similar to 2001 (Fig. 1). In all years, however, the crop was well supplied by irrigation as required. We hypothesize that higher yields in 2003 relative to those of 2001 and 2002 may have resulted from cooler temperatures during the seed-fill period (data not shown), which lengthened seed-fill duration (R5 to R6) (Table 1). For example, seed-fill duration of MG 0 soybean in 2003 was 18 and 13 d longer than in 2001 and 2002, respectively. Furthermore, the low temperatures throughout the growing season in 2003 resulted in an average of 11 and 6 d longer duration in time from emergence to R6 than in 2001 and 2002, respectively. Associated with the longer seed-fill period in 2003 was an increased average seed mass (data not shown). Average seed mass in 2003 over all genotypes and population densities was 162 mg seed^{-1} compared with 135 mg seed^{-1} in 2001 and 115 mg seed^{-1} in 2002. Previous research has noted that moderate day/night temperatures ($24/19^\circ\text{C}$) prolong seed fill and increase average seed mass relative to higher day/night temperatures ($33/28^\circ\text{C}$) (Egli and Wardlaw, 1980).

Data from 2002 indicate that cultivar selection within a MG played a significant role in asymptotic yield potential (Table 2). For example, the cultivars Trail, Dwight, and MPV437 had 27, 14, and 16% lower asymptotic yield potential than their counterparts within their respective

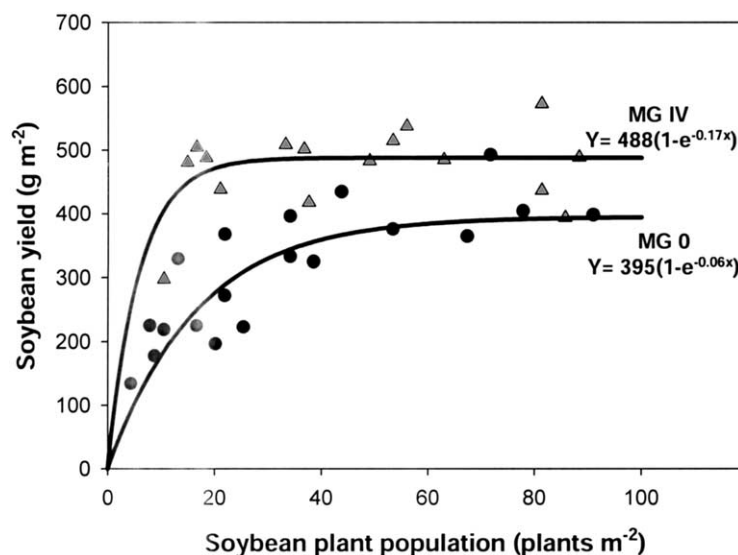


Fig. 2. Response of soybean yield to plant population for MG 0 and IV soybean in Fayetteville, AR, in 2003.

MG. With the exception of Trail, however, these data also indicate that cultivars which were used all three years of the experiment were well-adapted to the Mid-south soybean production area and, therefore, good indicators of how other well-adapted cultivars within their respective MG should perform (Edwards et al., 2003).

While generally not as large as for MG 00 and 0 soybean, yield responses to increased plant population for MG I and II cultivars were observed (Table 2). Cultivars from MG III and IV were variable in their response to increased plant population, and response to increased plant population was greatly affected by cultivar and environment. In 2001, for example, MG III and IV soybean displayed a marked response to increased plant population with β values of 0.09 and 0.13, respectively. In 2002, however, there was no response of yield to increased plant population in Macon, Pana, or Pioneer 94B01 soybean. Ball et al. (2000) also found differences in the yield responses between MG IV cultivars to increased plant population (i.e., estimation of β was nonsignificant), with determinate cultivar Manokin requiring a lower plant population to reach maximum yield than the indeterminate cultivar A4922. Branching, leaf angle and size, and other canopy architectural features likely affect a given cultivar's response to population, regardless of MG.

Cultivars from MG V and VI had similar maximum yield potential as did well-adapted MG III and IV cultivars in 2001 and 2002. In 2003, asymptotic yield was

similar for MG I through VI cultivars (Table 2). In contrast to earlier MGs, however, MG V and VI soybean generally had no yield response to increased plant population (i.e., a very large β or one with a confidence interval including zero).

It is also important to note that yield did not decrease at our highest population densities. No lodging was evident in our environments but may be a concern in some locations (e.g., Cooper, 1971). The most severe lodging occurred in our highest populations and latest MGs, but even these lodging scores did not exceed 2.5 on a 1 (no lodging) to 5 (severe lodging) scale (data not shown).

From the regression coefficients in Table 2, one may calculate a population density required to achieve 95% of the asymptotic yield. For example, in 2001 both Lambert (MG 0) and Hutcheson (MG V) had identical asymptotic yields (251 g m^{-2}), but to reach 95% of the asymptotic yield required 43 plants m^{-2} for Lambert and 16 plants m^{-2} for Hutcheson. Population densities greater than these values would not likely give yield increases, but higher populations could be used to lessen the risk of replanting if conditions were less than desirable for soybean emergence. The decision to replant for a particular MG would be governed by economic considerations involving yield response to population, seeding costs, irrigation, seasonal price fluctuations, and other economic factors (M. Popp et al., unpublished data, 2005).

Table 2. Coefficient estimates, standard errors, and significance level for the equation $Y = \alpha(1 - e^{-\beta x})$ describing soybean yield (g m^{-2} , $n = 20$) as a function of soybean plant population (plants m^{-2}) for maturity group 00 through VI soybean at Fayetteville, AR, in 2001, 2002, and 2003.

Year	Maturity group	Cultivar	α	β	Regression	R^2
2001	00	Trail	230 $\pm$ 14	0.05 $\pm$ 0.01	***	0.97
	0	Lambert	251 $\pm$ 17	0.07 $\pm$ 0.02	***	0.96
	I	IA 1006	228 $\pm$ 7	0.12 $\pm$ 0.02	***	0.99
	II	IA 2008	245 $\pm$ 12	0.13 $\pm$ 0.04	***	0.97
	III	Macon	273 $\pm$ 14	0.09 $\pm$ 0.02	***	0.98
	IV	Pioneer 94B01	267 $\pm$ 13	0.13 $\pm$ 0.04	***	0.96
	V	Hutcheson	251 $\pm$ 17	0.19 $\pm$ 0.07	***	0.96
	VI	NK 6262	265 $\pm$ 15	—	NS†	—
	00	Jim	161 $\pm$ 22	0.07 $\pm$ 0.04	***	0.86
	00	Trail	117 $\pm$ 17	0.09 $\pm$ 0.06	***	0.81
2002	0	AC Comoran	183 $\pm$ 29	0.04 $\pm$ 0.02	***	0.85
	0	Lambert	175 $\pm$ 19	0.08 $\pm$ 0.03	***	0.89
	I	IA 1006	238 $\pm$ 17	0.09 $\pm$ 0.03	***	0.95
	I	MN 1801	215 $\pm$ 13	0.17 $\pm$ 0.06	***	0.96
	II	Dwight	191 $\pm$ 12	0.15 $\pm$ 0.05	***	0.96
	II	IA 2008	220 $\pm$ 16	0.15 $\pm$ 0.05	***	0.94
	III	Macon	236 $\pm$ 18	—	NS	—
	III	Pana	231 $\pm$ 18	—	NS	—
	IV	Pioneer 94B01	261 $\pm$ 19	—	NS	—
	IV	MPV 437	220 $\pm$ 17	0.14 $\pm$ 0.06	***	0.93
	V	Caviness	256 $\pm$ 14	—	NS	—
	V	Hutcheson	246 $\pm$ 17	—	NS	—
	VI	NK 6262	243 $\pm$ 15	—	NS	—
	VI	Desha	253 $\pm$ 12	—	NS	—
	00	Trail	345 $\pm$ 95	0.03 $\pm$ 0.01	***	0.89
2003	0	Lambert	395 $\pm$ 29	0.06 $\pm$ 0.01	***	0.96
	I	IA 1006	423 $\pm$ 17	0.14 $\pm$ 0.04	***	0.98
	II	IA 2008	449 $\pm$ 16	0.15 $\pm$ 0.03	***	0.99
	III	Macon	481 $\pm$ 21	0.23 $\pm$ 0.10	***	0.98
	IV	Pioneer 94B01	488 $\pm$ 22	0.17 $\pm$ 0.07	***	0.98
	V	Hutcheson	434 $\pm$ 12	—	NS	—
	VI	NK 6262	470 $\pm$ 14	—	NS	—

*** Indicates the overall regression equation was statistically significant at the 0.001 level of significance.

† Data failed to converge for β after the maximum number of iterations had been exceeded or confidence the interval of β included zero, indicating there was no significant response to increased plant population.

Table 3. Analysis of variance table and Type III hypothesis tests for covariate analysis of soybean harvest index as a function of maturity group (MG) and plant population (PP) in 2001, 2002, and 2003.

Year	Source	DF	SS	MS	F value	Prob F	cv	R ²	
2001	Model	16	26.00	1.625	1178	<0.001	9	0.71	
	Error	123	0.17	0.001					
	Total	139	26.17						
Type III Hypothesis Tests									
2002	MG	8	8.80	1.100	797	<0.001	10	0.49	
	MG × PP	8	0.07	0.009	6	<0.001			
	Model	32	56.39	3.522	2076	<0.001			
	Error	281	0.50	0.002					
	Total	313	56.86						
Type III Hypothesis Tests									
2003	MG†	16	17.45	1.090	652	<0.001	7	0.51	
	MG × PP	16	0.11	0.007	4	<0.001			
	Model	16	30.78	1.924	2107	<0.001			
	Error	135	0.12	<0.001					
	Total	151	30.90						
Type III Hypothesis Tests									
	MG	8	9.10	1.137	1246	<0.001			
	MG × PP	8	0.03	0.004	5	<0.001			

† Cultivar nested within MG was nonsignificant ($P = 0.29$).

Harvest Index and Height of First Pod

Harvest index of MG 00 and 0 soybean increased with increasing plant population, but, with the exception of MG III soybean in 2002, HI of MG I, II, and III soybean had no response to increased plant population (Tables 3 and 4). Harvest index of MG IV, V and VI soybean decreased as plant population increased, with MG IV soybean in 2002 being the exception. Higher HI values are usually associated with early-maturing cultivars relative to later-maturing cultivars (Johnson and Major, 1979; Schapaugh and Wilcox, 1980). Using regression coefficients from Table 4, predicted HI values at popula-

Table 4. Coefficients, standard errors, and P values for covariate analysis using the model $Y = \beta_0 + \beta_1 x$ describing harvest index of maturity group 00 through VI soybean cultivars as a function of plant population at Fayetteville, AR, in 2001, 2002, and 2003.

Year	Maturity group	Cultivar	β_0	β_1
2001	00	Trail	0.49 ± 0.02**	0.0002 ± 0.0002NS
	0	Lambert	0.49 ± 0.02**	0.0003 ± 0.0002NS
	I	IA 1006	0.40 ± 0.02**	-0.0002 ± 0.0002NS
	II	IA 2008	0.45 ± 0.02**	-0.0005 ± 0.0002NS
	III	Macon	0.43 ± 0.02**	-0.0005 ± 0.0003NS
	IV	Pioneer 94B01	0.44 ± 0.02**	-0.0007 ± 0.0003NS
	V	Hutcheson	0.46 ± 0.02**	-0.0028 ± 0.0005**
2002	VI	NK 6262	0.42 ± 0.02**	-0.0010 ± 0.0003**
	00	—†	0.44 ± 0.0**	0.0006 ± 0.0002*
	0	—	0.42 ± 0.01**	0.0007 ± 0.0003*
	I	—	0.40 ± 0.01**	0.0004 ± 0.0002*
	II	—	0.38 ± 0.01**	0.0002 ± 0.0003NS
	III	—	0.44 ± 0.01**	-0.0001 ± 0.0002NS
	IV	—	0.46 ± 0.01**	-0.0001 ± 0.0002NS
2003	V	—	0.44 ± 0.01**	-0.0001 ± 0.0002**
	VI	—	0.41 ± 0.01**	-0.0008 ± 0.0002**
	00	Trail	0.42 ± 0.01**	0.0014 ± 0.0003**
	0	Lambert	0.47 ± 0.01**	0.0005 ± 0.0003NS
	I	IA 1006	0.45 ± 0.01**	-0.0001 ± 0.0002NS
	II	IA 2008	0.46 ± 0.01**	-0.0003 ± 0.0003NS
	III	Macon	0.47 ± 0.01**	-0.0001 ± 0.0003NS
	IV	Pioneer 94B01	0.49 ± 0.02**	-0.0007 ± 0.0003**
	V	Hutcheson	0.45 ± 0.01**	-0.0005 ± 0.0002*
	VI	NK 6262	0.43 ± 0.01**	-0.0006 ± 0.0002**

* Significant at the 0.05 level.

** Significant at the 0.01 level.

† Cultivar nested within MG was nonsignificant in predicting harvest index in 2002; therefore, data have been averaged by MG.

tions of ≥ 40 plants m^{-2} follow this general axiom that HI decreases as MG increases. In a companion manuscript (Edwards et al., 2005a) we noted that by combining data for MGs, years, and plant populations that HI decreased linearly as a function of the cumulative amount of light intercepted from emergence to R6.

A typical harvest cutting height for soybean is between 7.5 and 12.5 cm above the soil surface (Grabau and Pfeiffer, 1990); therefore, if the first fertile node is below this harvest height, significant harvest losses can occur. Equation [1] explained the relationship between height of first fertile node and plant population very well with R^2 values ranging between 0.83 and 0.99 (Table 5).

Table 5. Coefficient estimates, standard errors, and significance level for the equation $Y = \alpha(1 - e^{-\beta x})$ describing average height of first fertile node (cm, $n = 20$) as a function of soybean plant population (plants m^{-2}) for maturity group 00 through VI soybean at Fayetteville, AR, in 2001, 2002, and 2003.

Year	Maturity group	α	β	Regression	R ²
2001	00	16 ± 3.2	0.01 ± 0.004	***	0.97
	0	11 ± 0.8	0.03 ± 0.006	***	0.98
	I	13 ± 0.6	0.03 ± 0.004	***	0.99
	II	13 ± 0.8	0.05 ± 0.010	***	0.97
	III	17 ± 1.8	0.03 ± 0.006	***	0.97
	IV	23 ± 2.2	0.03 ± 0.005	***	0.97
2002	V	17 ± 1.6	0.1 ± 0.036	***	0.95
	VI	21 ± 1.9	0.04 ± 0.008	***	0.97
	00	5 ± 0.3	0.11 ± 0.032	***	0.99
	0	7 ± 0.6	0.05 ± 0.016	***	0.92
	I	10 ± 0.7	0.06 ± 0.013	***	0.93
	II	8 ± 1.2	0.06 ± 0.023	***	0.83
2003	III	13 ± 0.7	0.06 ± 0.012	***	0.94
	IV	12 ± 0.6	0.11 ± 0.024	***	0.94
	V	24 ± 1.8	0.12 ± 0.045	***	0.93
	VI	31 ± 2.2	0.08 ± 0.025	***	0.95
	00	8 ± 0.8	0.04 ± 0.010	***	0.97
	0	13 ± 1.6	0.03 ± 0.008	***	0.95
	I	18 ± 1.1	0.05 ± 0.009	***	0.97
	II	15 ± 0.9	0.08 ± 0.016	***	0.97
	III	19 ± 0.8	0.07 ± 0.013	***	0.98
	IV	21 ± 1.5	0.06 ± 0.014	***	0.97
	V	28 ± 1.5	0.08 ± 0.017	***	0.97
	VI	27 ± 1.5	0.07 ± 0.015	NS†	—

*** Significant at the 0.001 level.

† Regression equation was nonsignificant at the 0.05 level of significance.

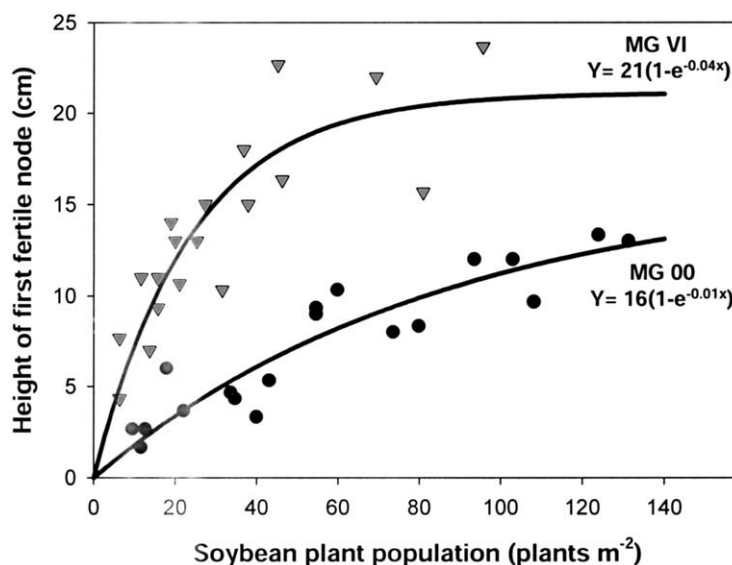


Fig. 3. Response of height of first fertile node to soybean plant population for MG 00 and VI soybean at Fayetteville, AR, in 2001.

Pairwise analysis of regression coefficients within a MG indicated that data could not be combined over years of the experiment. Height of first fertile node generally increased as soybean plant population increased (Fig. 3). Similar fits were obtained for all MGs each year of the experiment; therefore, only regression coefficients, their standard errors, and overall significance levels are presented (Table 5). With the exception of MG VI soybean in 2003, distance from the soil surface to first fertile node increased as plant population increased. However, response to increased plant population was variable, and the greatest response (smallest β coefficient) was generally in the earliest MG. Yield data indicate that plant population densities of MG II and earlier soybean necessary to increase the height of the first fertile node to a harvestable height of approximately 10 cm would also be required for highest yield potential in these MGs. Therefore, from a crop production standpoint, any concern over placement of the lowest pod would be a moot point, as seeding densities required to maximize yield potential also result in sufficient height of first fertile node.

CONCLUSIONS

These data show that under irrigated conditions MG II through VI soybean have similar yield potential in the Midsouth, but MG II, III, and IV soybean generally have lower irrigation requirements than later MGs. Maturity group I soybean had slightly lower asymptotic yield potential than later soybean in this experiment and required much higher seeding density to maximize yield. Irrigation requirements for MG I and II cultivars, however, were much lower than for later MGs. Different irrigation requirements among years for the same MG demonstrate the dependency of irrigation on crop phenology and timeliness of rainfall. An economic analysis by M. Popp et al. (unpublished data, 2005) considers if irrigation savings and seasonal price fluctuations of MG

I and II cultivars are sufficient to offset the higher seeding cost necessary to maximize their yield.

The differing maximum yield potential among years of this experiment exemplifies the problems associated with attempting to represent soybean yield simply as a function of plant population. Using this method to determine the agronomic optimal plant population necessitates averaging of data from years or locations that may or may not be statistically similar. Not only do population responses differ among MGs at a given location, as shown in these experiments, but they also differ within a MG and location for different sowing dates (Ball et al., 2000). A model that incorporates the underlying factors that determine the relationship between yield and plant population, therefore, would be beneficial in developing broader-based recommendations on seeded population. In a subsequent paper (Edwards et al., 2005a), we consider that yield response is a function of cumulative radiation interception which is affected by plant population and crop maturity.

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